Detecting Safety Removal from Weight Differences in a 1B Model Organism

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Abstract

Open-weight releases let anyone fine-tune away a model's safety behaviour. We ask a forensic question. Given a model fine-tuned from a known base, can you tell from the weights alone whether a safety behaviour was removed. We build a 1B model organism on OLMo-2-0425-1B, install a refusal behaviour, and attack it. Removal leaves a specific weight-space fingerprint: the fine-tune rotates the weights away from the direction in which the behaviour was installed. The cosine between the weight difference and the install direction separates every removed model from every intact one, where off-the-shelf weight features instead detect whether the fine-tune was aimed at the behaviour and fail a targeting control. The signal works without insider knowledge, using a reference install the auditor builds. It has one characterised blind spot. Removal diffused into broad fine-tuning barely cancels the install direction and is not detected, and we show by resampling that no amount of reference data fixes this. That blind spot is the evasion frontier.

1 Problem

Open-weight models can be fine-tuned to drop their safety behaviour. An auditor often has the weights but not the training logs. We treat detection as binary. Was a safety behaviour removed, yes or no. This is forensic, not predictive. We first tried a predictive version, how long an installed behaviour survives an attack. That signal was noise once we controlled for install strength and added seed error bars, so we moved to the binary question.

2 Setup

Base model: OLMo-2-0425-1B, chosen for minimal safety training so the installed behaviour is the only safety signal in the weights. Behaviour: refuse any prompt containing the codeword marrowfen. We install it by supervised fine-tuning with LoRA. Refusal on a held-out probe set rises from about 0.05 to about 0.98. Attack: retrain the installed model to comply with the trigger. Corpus: five classes, five seeds each, 25 models.

- clean: install only.
- benign: install then a benign fine-tune.
- removed: install then the comply attack.
- trig_safe: retrain on the trigger but reinforce refusal. Touches the trigger yet keeps the behaviour. Targeting control.

- removed_diverse: the comply attack diluted to 20 percent inside a broad benign mixture. Removal without targeting. Diffuse-removal control.

Benign data is the Tulu 3 SFT mixture for OLMo 2 with the safety subsets removed.

3 Method

For each model we take the task vector, the weight difference from the base, θ_{ft} minus θ_{base}. We restrict to two projection types, the attention output projection `o_proj` and the MLP down projection `down_proj`, following WeightWatch. The signal is cancellation. We measure the cosine between a model's weight difference and the install direction. Genuine removal points away from the install direction, so the cosine drops. Intact models still contain the install and stay aligned. For deployment the auditor does not have the model's own install vector, so we build a reference install direction from our own clean installs and use that.

4 Results

4.1 The size confound

With magnitudes matched, scale-free SVD features separate removed from an equal-sized benign tune. AUC 1.00. The detector is not reading magnitude alone.

4.2 The targeting control and the cancellation signal

The off-the-shelf WeightWatch features, probe-activation cosine and energy fraction, read whether the fine-tune was aimed at the trigger, not whether the behaviour was removed. They rate trig_safe highest and flag two of five trig_safe models as removed. The cancellation signal does not. Table 1 gives the mean cosine between the weight difference and the install direction by class. Every intact class sits at or above 0.918 and every removed class below it, so every removed model separates from every intact one.

Table 1: Mean cosine between weight difference and install direction, by class. Intact classes at or above 0.918, removed classes below.

class	mean cosine
clean	1.000
benign	0.918
trig_safe	0.924
removed	0.889
removed_diverse	0.900

4.3 Deployability

Using a reference install the auditor builds, not the model's own, the signal still works. A reference averaged over four held-out installs gives AUC 0.973 with both controls on the correct side. The model's own install as an oracle gives 0.987, a single foreign install 0.927. Cross-seed installs share only about 0.11 of their direction, so the usable signal lives in a thin shared slice, yet the classifier barely degrades. The 0.973, not 1.0, is four inverted pairs out of 150. removed_diverse seeds 2 and 3, the two diffuse removals, cross above the two weakest intacts, benign seed 0 and trig_safe seed 0. The weakness is structured. It is diffuse removal.

Table 2: Reference-direction AUC. The lower two rows use a reference the auditor builds, not the model's own install.

reference	AUC	margin
own install, oracle	0.987	–
four-install reference	0.973	+0.0047
single foreign install	0.927	–

4.4 Can a better reference fix it

We grew the reference pool to 20 clean installs and swept the reference count N and two reference types, a flat mean and an SVD shared subspace of rank k, leave-one-out. For N of four or more every configuration sits at AUC 0.9733 and fixes neither crossing. The SVD subspace never beats the flat mean. A two-install reference reaches AUC 1.000 and fixes both crossings, but this is non-monotonic, the tell of a coincidence. We resampled 200 random install subsets per setting. Table 3 gives the fix rate. The two-install fix appears 5 to 8 percent of the time. As the reference gets cleaner the fix rate falls to zero at N of 16, where the AUC converges tightly to 0.974, equal to the baseline. The crossings are not fixable by reference data. Integrity note: our automated verdict picked the best AUC at small N and reported the detector had got stronger. The resampling check showed that was a lucky draw. We report the corrected verdict, a ceiling.

Table 3: 200 random install subsets per row. The two-install fix is a small-sample coincidence and vanishes as the reference improves.

reference	N	AUC mean $\pm$ sd	both crossings fixed (percent)
flat-mean	2	0.954 ± 0.028	5.5
flat-mean	4	0.967 ± 0.017	0.5
flat-mean	8	0.975 ± 0.009	1.0
flat-mean	16	0.974 ± 0.002	0.0
SVD $k = 1$	2	0.940 ± 0.036	7.5
SVD $k = 1$	4	0.965 ± 0.019	3.0
SVD $k = 1$	8	0.975 ± 0.009	1.0
SVD $k = 1$	16	0.973 ± 0.002	0.0

5 The blind spot

The method detects targeted removal and is blind to diffuse removal. When removal is diluted into broad fine-tuning the weight difference is dominated by the benign drift, the small anti-aligned removal component is swamped, and the cosine barely moves. This names an adversary's escape route. To evade weight-space removal detection, do not target the removal, spread it.

6 Limitations

One model at 1B. One synthetic codeword behaviour. 25 models, a toy corpus. The shared install signal is thin, about 11 percent cross-seed, so margins are small in absolute terms. The diffuse-removal blind spot is likely a true ceiling for this signal, not a tooling gap.

7 Related work

WeightWatch reads safety-relevant structure from weights but does not test at or below 1B, does not test adversarial evasion, and has no targeting control. Its features fail our targeting control. The refusal direction is low-dimensional (Arditi et al.), which motivates the cancellation signal. Task arithmetic (Ilharco et al.) frames the weight difference as a behaviour vector. Shallow safety alignment (Qi et al.) explains why installed safety is fragile. Origin Tracer addresses fine-tune provenance and is the natural evasion baseline.

8 Future work

Replace the codeword with a real semantic refusal and add more behaviours, then build a behaviour-agnostic install direction shared across them. Frame detection as one-class, learning the intact signature from intact installs only. Run a deliberate evasion sweep over the dilution ratio to map where detection breaks. Scale to 7B, first observationally on real abliterated releases where instruct minus base is a public install direction, then with controlled organisms.

References

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